

TEAM T-800



AGENTIC COMPLIANCE PLATFORM

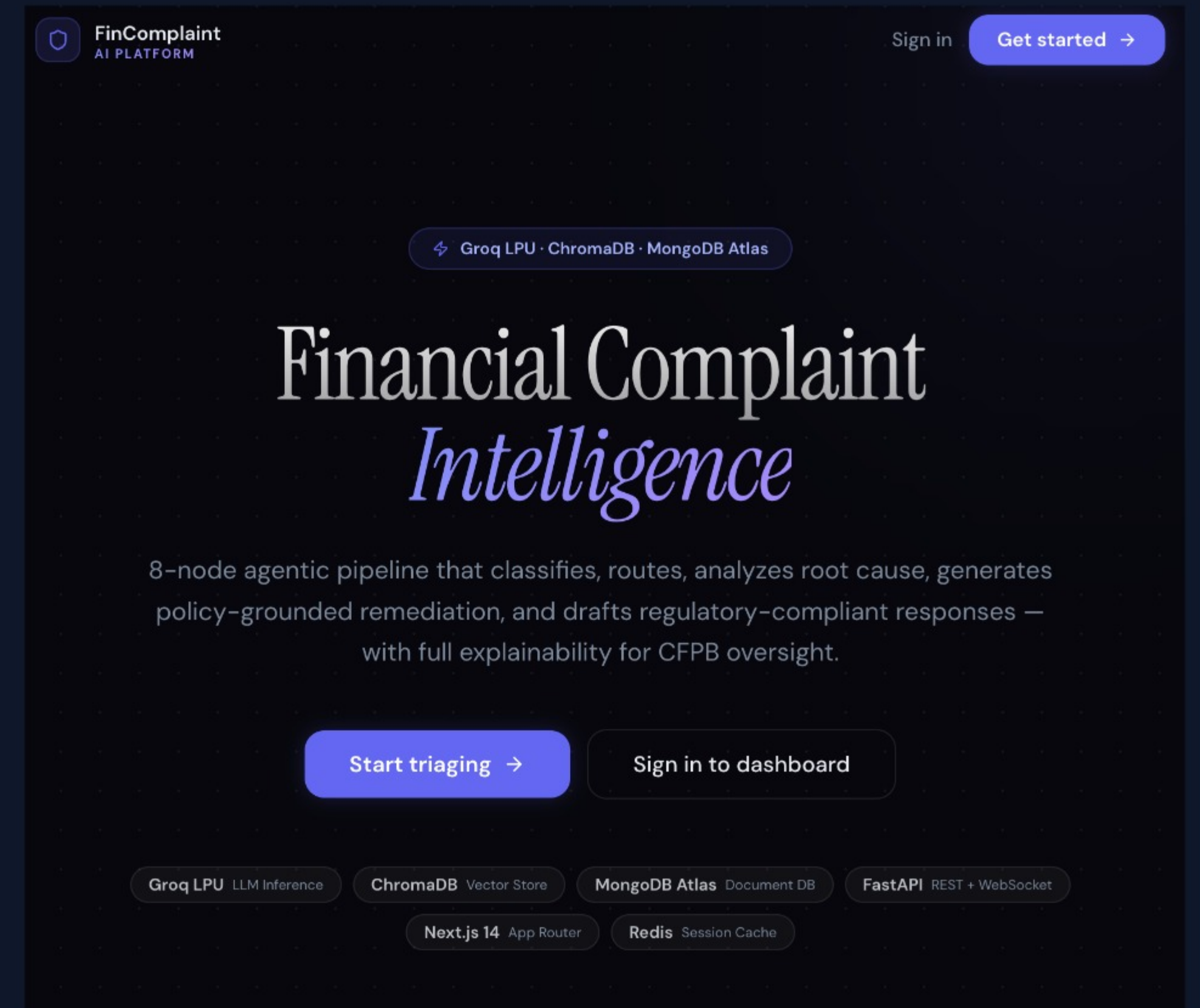
FinComplaint AI

Intelligent Consumer Complaint Resolution System

🤖 8 Specialized Agents

⚡ CFPB Compliant

🔒 PII-Safe Pipeline

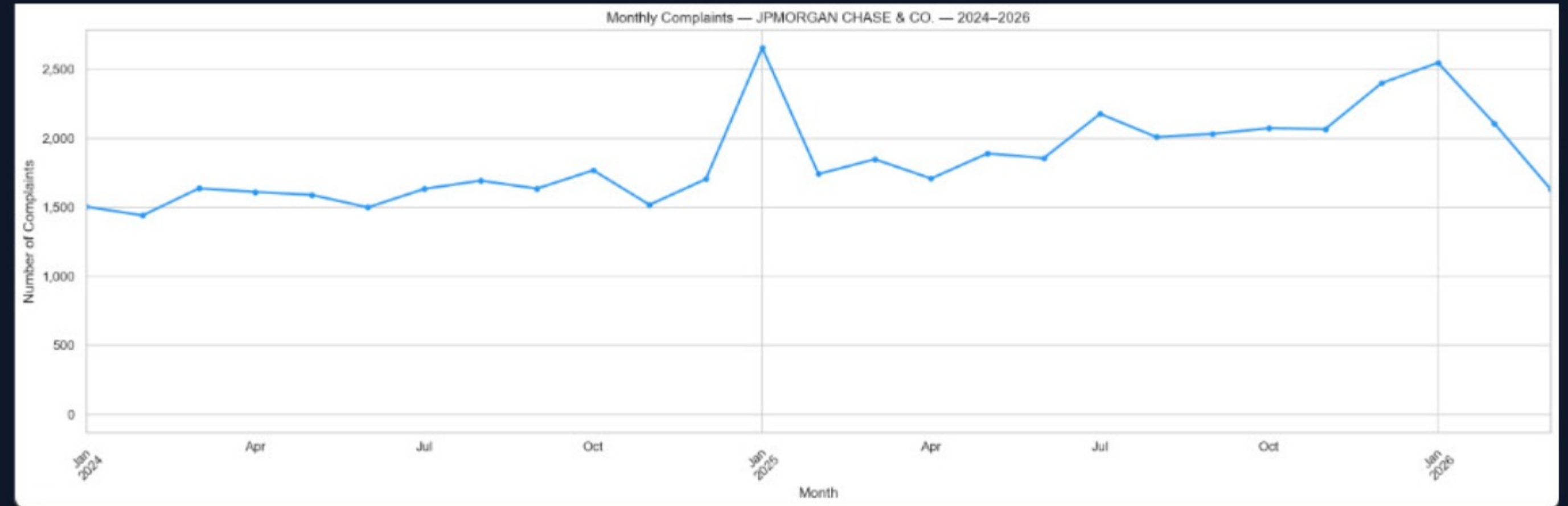


What Data tells ?

CFPB Dataset

Database of financial product and service complaints (2024-2026)

JP MORGAN CHASE AND CO.



- I CALLED !
- Sensitive information
- Disputes
- Products : transaction, account, card, payment

Why FinComplaintAI ?

 CFPB Scrutiny Is Intensifying

Financial complaints face unprecedented regulatory oversight and enforcement action

 Slow Human Review Creates Risk

Missed SLA deadlines result in regulatory penalties and reputational damage

 Manual Processes Can't Scale

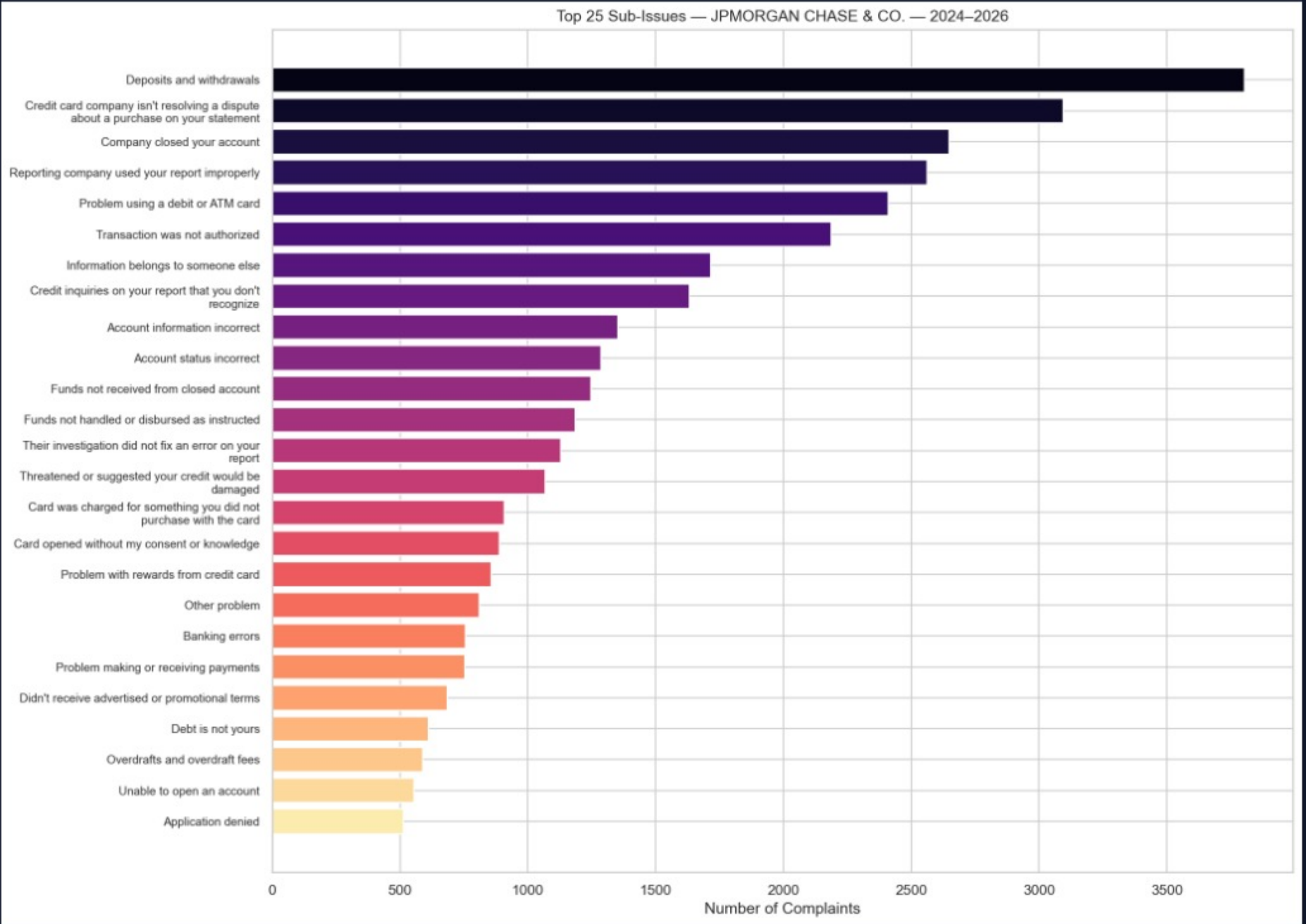
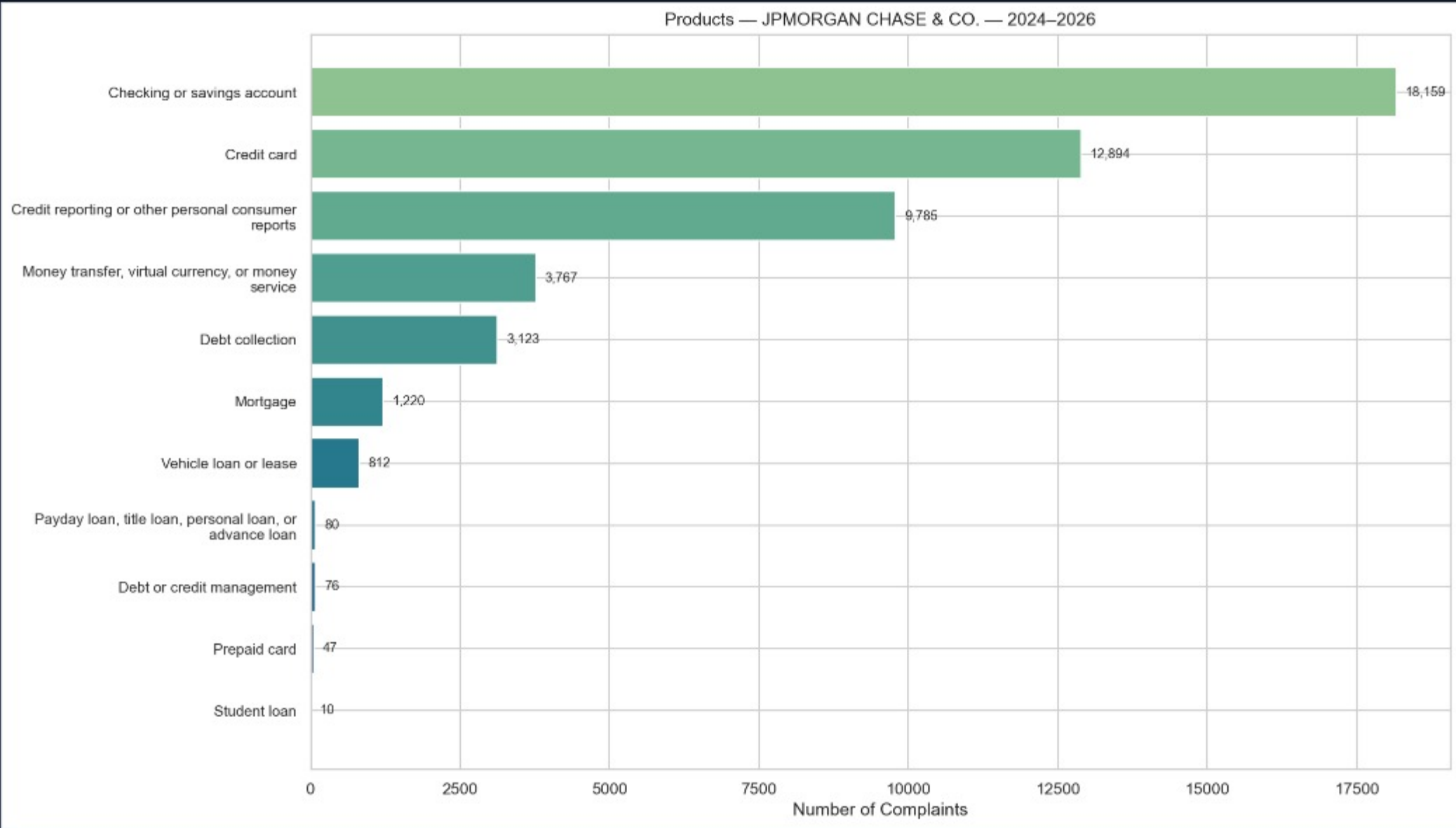
Complaint volume growth outpaces staffing, quality degrades under load

 PII/PCI Data Vulnerabilities

Sensitive customer data exposed across fragmented toolchains and workflows

 Inconsistent State SLA Compliance

50-state regulatory patchwork makes consistent compliance nearly impossible



Business Impact

Measurable value across compliance, scale, security, and auditability



Compliance Automation

Deep MCP policy integration ensures every response meets state-specific regulatory requirements, reducing human error in compliance workflows



Scale Without Risk

90% of routine complaints resolved automatically. Only the top 10% high-risk cases escalate to human reviewers for critical judgment calls



Data Privacy

Preemptive PII scrubbing at the pipeline entry point ensures LLMs never see restricted customer data — privacy by architecture, not policy



Verifiable Audit Trails

Granular SQLite logging captures every node-level decision, eliminating the "black box" problem that regulators flag in AI systems

From Complaint to Resolution

Fraud case walkthrough — unauthorized transfer with exposed SSN

Complaint

A debt collector is demanding \$2,400 for a store credit card I never opened. I have never had an account with this retailer and the address they have on file is not mine. I sent a written debt validation request 40 days ago but they have not responded and are still reporting this fraudulent debt to the credit bureaus.

Customer Response

We understand the concern described in your complaint and appreciate the chance to review it. Current findings point to Debt collector's failure to respond to debt validation request, resulting in continued reporting of fraudulent debt to credit bureaus. We will follow up within 21 calendar days with the next status update or any additional information needed.

✓	Intake & PII Scrubbing	
	PII redacted · text normalized	
✓	Complaint Classification	2265ms
	1314 tokens	
✓	Routing Decision	526ms
	Route: continue	
	1354 tokens	
✓	Root Cause Analysis	33373ms
	Debt collector's failure to respond to debt validation request, resulting in continued reporting of fraudulent debt to credit bure...	
	1302 tokens	

✓	Remediation Planning	57112ms
	Assigned: Debt Collection	
	766 tokens	
✓	Response Drafting	18553ms
	Response letter drafted	
	4325 tokens	
✓	Compliance Audit	16504ms
	✓ Compliance verdict: PASS	
	1904 tokens	
✓	Explanation Generation	49097ms
	Explanation chain generated	
	5931 tokens	

System Architecture

Purpose-built components, cleanly decoupled

Single-Purpose Agents

Groq-powered: Classifier, RootCause, Remediator, Writer, Auditor, Explainer

Graph & State Layers

LangGraph-inspired orchestration with deterministic state transitions

ChromaDB

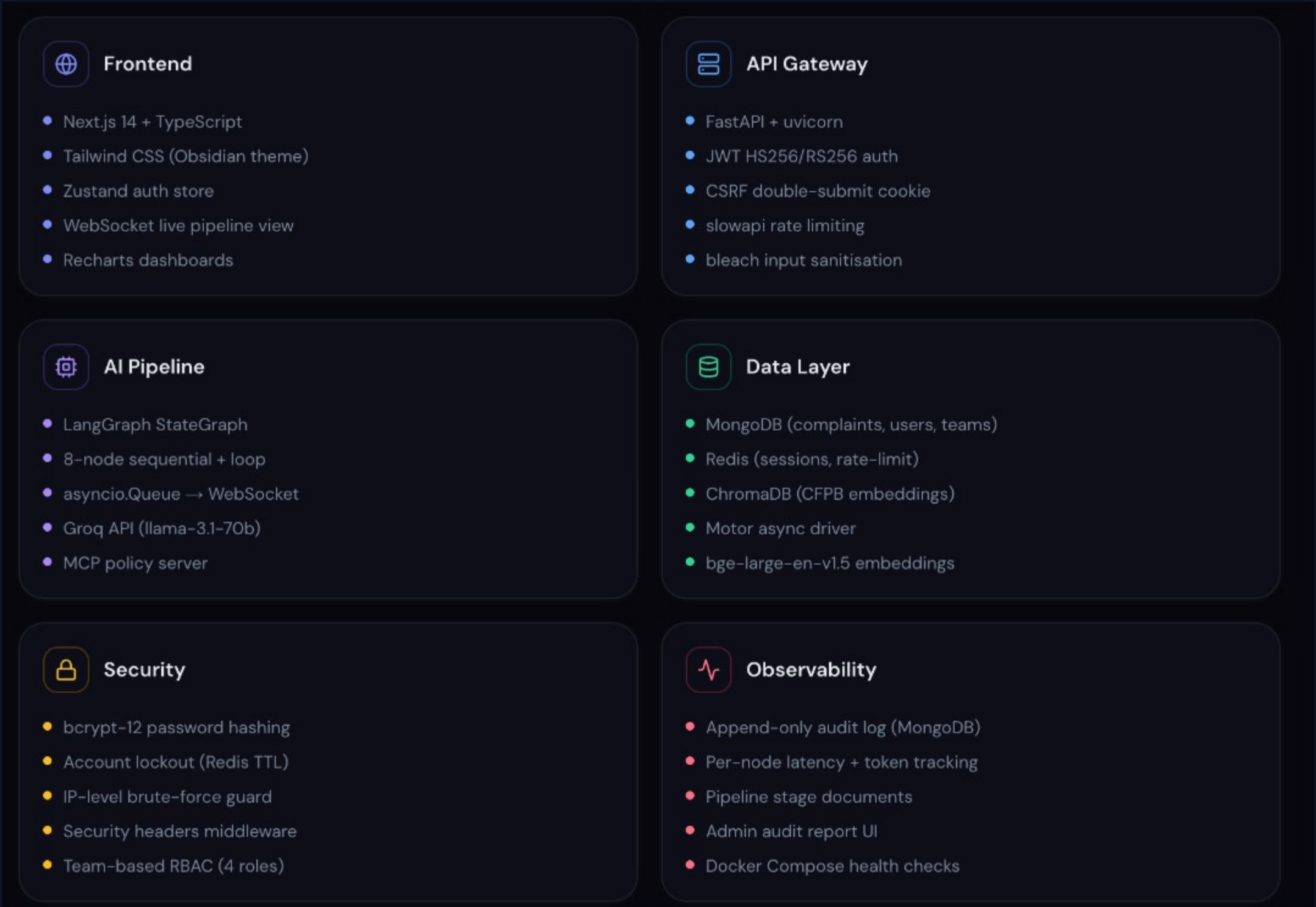
Local vector store for historical complaint pattern matching and root-cause analysis

MCP Policy Server

Decoupled regulatory logic — state SLAs, compliance rules, policy updates

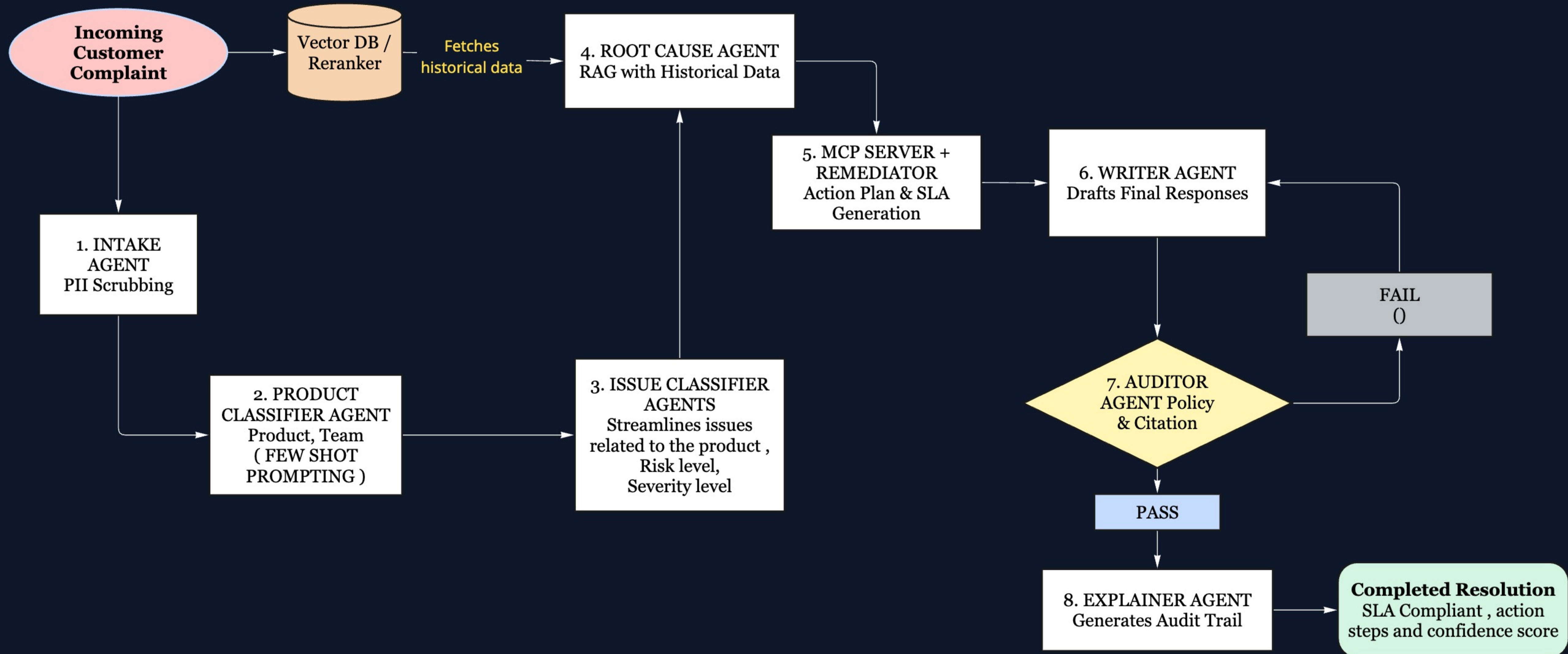
SQLite Audit Database

Node-level event tracking for complete decision transparency



End-to-End Workflow

EIGHT-stage agentic pipeline from intake to resolution



Engineering Highlights

Built for reliability, not just capability



Roll Based Access Control - Team and Admin Page



Bcrypt Hashing and Rate Limiting



JWT Authentication



Scaling



Observability



State-Driven Agentic Pipeline

Sophisticated graph-based orchestration ensures predictable, debuggable agent behavior at every node



Deterministic Fallback Heuristics

When LLMs fail or timeout, rule-based fallbacks maintain service continuity without compromising compliance



Human-in-the-Loop Gates

Critical and high-risk cases automatically pause for human manager review before proceeding



Writer-Auditor Loop with Cycle Caps

Iterative validation loop refines responses until compliant — with hard caps preventing infinite cycles

What we Achieved !!

1. Scaled for over 7 products and over 40 issues types
2. Handled 3000 historical complaints
3. Inference time under a minute
4. Monitoring per agent token usage for Optimisation.
5. Fallback Safe Integration
6. Performance metrics per Agent calculated - Product and Issue classifier achieved above 85%



THANKYOU !

